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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, NOV. - DEC. 2025**25ECE101 – APPLIED DIGITAL LOGIC DESIGN**

(For ACT, AER, BTY, CIV, ECE, EEE, MEC, VDT, AID, AIM, CYB, CCE, CSE, CBS, ISE, RAI)

Duration: 3 Hours]**[Max. Marks: 100****Note:**

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The Boolean expression $AB+AC'+BC$ simplifies to
A) $BC+AC'$ B) $AB+AC'+B$
C) $AB+AC'$ D) $AB+BC$
2. What are the canonical forms of Boolean expressions?
A) OR and XOR B) NOR and XNOR
C) MAX and MIN D) SOP and POS
3. An OR gate and a ----- gate will form the NOR gate.
A) OR B) NAND
C) AND D) NOT
4. If we add an inverter at the output of the AND gate, what function is produced?
A) XOR B) NAND
C) OR D) NOR
5. One operation that is not given by the magnitude comparator is
A) Equal B) Less than
C) Greater than D) Sum
6. Adding 1001 and 0010 gives the output of
A) 1011 B) 1111
C) 0000 D) 1010
7. The major difference between half-adders and full-adders is
A) No difference at all B) A full adder can take cin in addition to a and b
C) A half adder has more outputs D) A full adder has more outputs
8. Let the input of a subtractor be A and B, then what will the output be if $A = B$?
A) 0 B) 1
C) Equal D) Greater than B
9. If A and B are the inputs of a half adder, the sum is given by ____
A) $A \text{ xor } B$ B) $A \text{ and } B$
C) $A \text{ or } B$ D) $A \text{ xnor } B$
10. A 4x1 Mux will have
A) 4 outputs B) 1 input
C) 2 outputs D) 2 select lines
11. The encoder's job is to
A) Reduce the number of output bits B) Act like a decoder
C) Act like a mux D) Act like an adder
12. One way to make a four-bit adder to perform subtraction is by
A) Inverting the second input B) Inverting the carry in
C) Inverting both the inputs D) Not changing anything

13. A circuit that converts n inputs to 2^n outputs is called
 A) adder B) subtractor
 C) Full adder D) decoder
14. The characteristic equation for the J-K flip-flop is
 A) $J\bar{Q} + \bar{K}Q$ B) $JQ + KQ$
 C) $\bar{J}Q + K\bar{Q}$ D) $J\bar{Q}Q + \bar{K}Q$
15. What is meant by parallel-loading the register?
 A) Shifting the data in all flip-flops simultaneously B) Loading data into the first flip-flop
 C) Loading data in all four flip-flops at the same time D) Momentarily disabling the synchronous SET and RESET inputs
16. Which of the following statements are TRUE regarding shift registers?
 A) A shift register is a group of flip-flops B) It is not used for data storage
 C) It is not used for data movement D) It has no memory
17. For toggling the output, which flip-flop can be used?
 A) SR flip-flop B) SR latch
 C) RS latch D) T flip flop
18. Count sequence 0,2,4,6,0 is an example for —
 A) Down counter B) up/down counter
 C) Shift register D) Up counter
19. The minimum number of flip-flops that can be used to construct a modulus-5 counter is - -
 A) 3 B) 6
 C) 9 D) 5
20. Parallel In Parallel Out is —
 A) A counter circuit B) An encoder
 C) A mux D) A shift register

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I

	Marks	BT*	CO*	PO*
1. a) Convert the following: (i) $(B65F)_{16} = (?)_{10}$ (ii) $(110101)_2 = (?)_{10}$ (iii) $0.6875_{10} = (?)_2$	6	3	1	1
b) Draw the logic diagram using basic gates for the given Boolean equations. Also, simplify the following Boolean functions to a minimum number of literals. (i) $xy + x'z + yz$ (ii) $(x + y)(x + y')$	6	2	1	1
c) Write the truth table and switching equation for a model that accepts a 3-bit input and produces a 1-bit output, such that the output is high only if the first two bits are equal; otherwise, it is low.	4	3	1	1
2. a) State and prove DeMorgan's theorem using the tabular method. Also, find the complement of the functions using DeMorgan's theorem: $F1 = x'yz' + xyz$ $F2 = x(y'z' + yz)$	6	3	1	1
b) Write a truth table and a switching equation for a digital system whose output 1 is defined as logically high if the 4-bit input binary number is an even number; otherwise, the output will be logically low. The output 2 is defined if and only if the input binary number is greater than 9.	6	3	1	1
c) Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$ and (b) $Y - X$ by using the 2's complement method.	4	3	1	1

Unit – II

3. a) Use a K-map to simplify the following Boolean equations:
 (i) $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$
 (ii) $F = A'B'C' + B'CD' + A'BCD' + AB'C'$
 (iii) $F(w, x, y, z) = \Sigma M(1, 3, 7, 11, 15) + d(0, 2, 5)$ 6 3 2 1
 b) Design a 3-input digital circuit that gives a high output if the input binary number has more ones than zeros.
 (i) Write a truth table
 (ii) Write the equation in shorthand form
 (iii) Use a K-map to obtain the minimal sum.
 (iv) Draw the logic diagram for the minimal sum. 6 3 2 1
 c) Realize the given expression using logic gates:
 $F(w, x, y, z) = x' + y' + w'z'$ 4 3 2 1
4. a) Find the implicants of $F(A, B, C, D) = \Sigma m(0, 2, 3, 6, 7, 8, 10, 12, 13)$ using Quine Mccluskey Method. 6 3 2 1
 b) Simplify the following using 3-variable k-map:
 (i) $xy + x'y'z' + x'yz'$
 (ii) $x'y' + yz + x'yz'$ 6 3 2 1
 c) Realize the given expression using logic gates:
 $P(a, b, c) = a'b' + ba + (ba)'$ 4 3 2 1

Unit – III

5. a) With the help of a truth table and Boolean expressions, explain the working of:
 (i) 3 to 8 decoder
 (ii) 4x1 Mux 6 2 3 1
 b) Explain 4-bit parallel adder-subtractor design using Full Adders with suitable examples. 6 3 3 1
 c) Write the truth table for a 1-bit comparator. Obtain the switching equations for output variables and draw the logic diagram. 4 3 3 1
6. a) Implement the following using an 8x1 Mux:
 (i) $F(a, b, c, d) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$
 (ii) $F(a, b, c) = \Sigma(0, 4, 5, 6)$ 6 3 3 1
 b) Explain the design of a half adder and a full adder using logic gates. 6 2 3 1
 c) Implement the following functions using a decoder minimizing the number of inputs to be summed:
 (i) $f_1(a, b, c) = \Sigma(0, 2, 3, 5, 6, 7)$
 (ii) $f_2(a, b, c) = \Sigma(1, 3, 4, 6, 7)$ 4 3 3 1

Unit – IV

7. a) Obtain the characteristic equation of:
 (i) JK flip-flop
 (ii) T flip-flop 6 3 4 1
 b) Explain the working of the SR latch built using NOR gates. 6 2 4 1
 c) Explain the difference between a latch and flip flop. 4 2 4 1
8. a) Explain the operation of the Master Slave JK flip-flop with a truth table. Write the symbol for the same. 6 2 4 1
 b) Show how to convert a JK flip-flop to a D flip-flop. 6 2 4 1

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| c) With the help of a block diagram, write the difference between combinational and sequential circuits. Give two examples for each. | 4 | 2 | 4 | 1 |
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Unit – V

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| 9. a) Design a synchronous 4-bit up counter using D flip-flops. | 10 | 3 | 5 | 1 |
| b) Explain the design and working of a parallel in parallel out operation in a shift register using D flip-flops. | 6 | 2 | 5 | 1 |
| 10. a) Design a synchronous 3-bit down counter using positive-edge-triggered T flip-flops with minimal combinational gating. | 10 | 3 | 5 | 1 |
| b) Explain the design of a serial in serial-out shift register using D flip-flops. | 6 | 2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
